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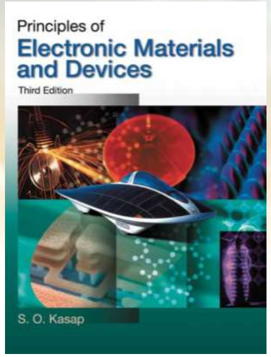

《固态电子学》
Solid-State Electronics

Ch4-2
Modern Theory of Solids

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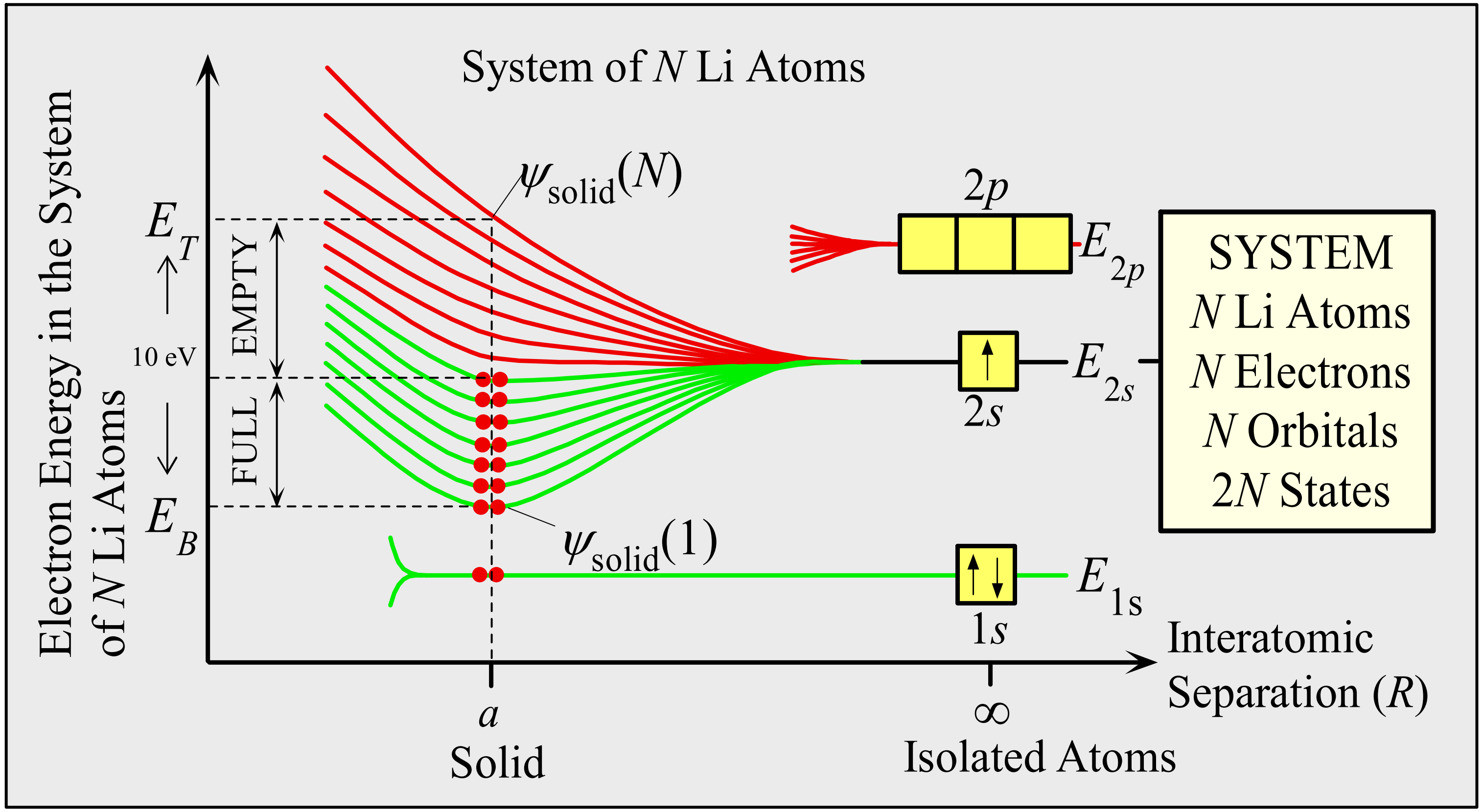
Some slides are modified from other Profs' PPTs. Their contributions are greatly acknowledged.

The Case of Solid

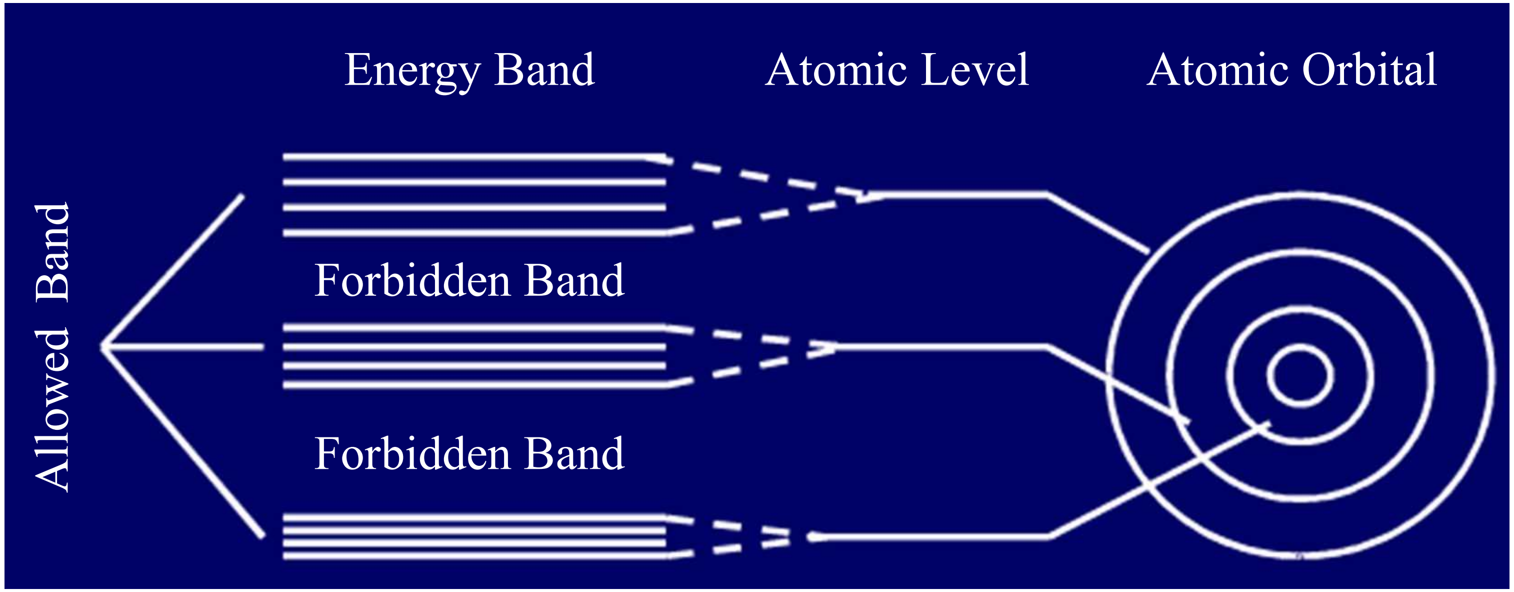


- As N Li atoms approach together, each 2s electron has different potential energy, depending on their position.
- Resulting in N different wave function. Hence the energy level of 2s splits into N finely separated energy levels.
- There are N 2s electrons, but 2N states. Electrons sit the orbital with lower energy first.

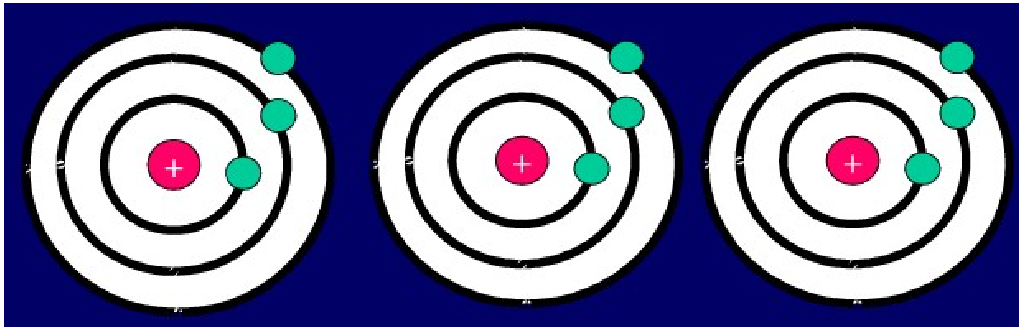
Formation of energy band

When N atoms to form a crystal

- Allowed Band: Each N-degree degenerate energy level is split into energy levels that are close to each other. These N energy levels form an energy band.
- Forbidden Band: A band with no energy level between the bands

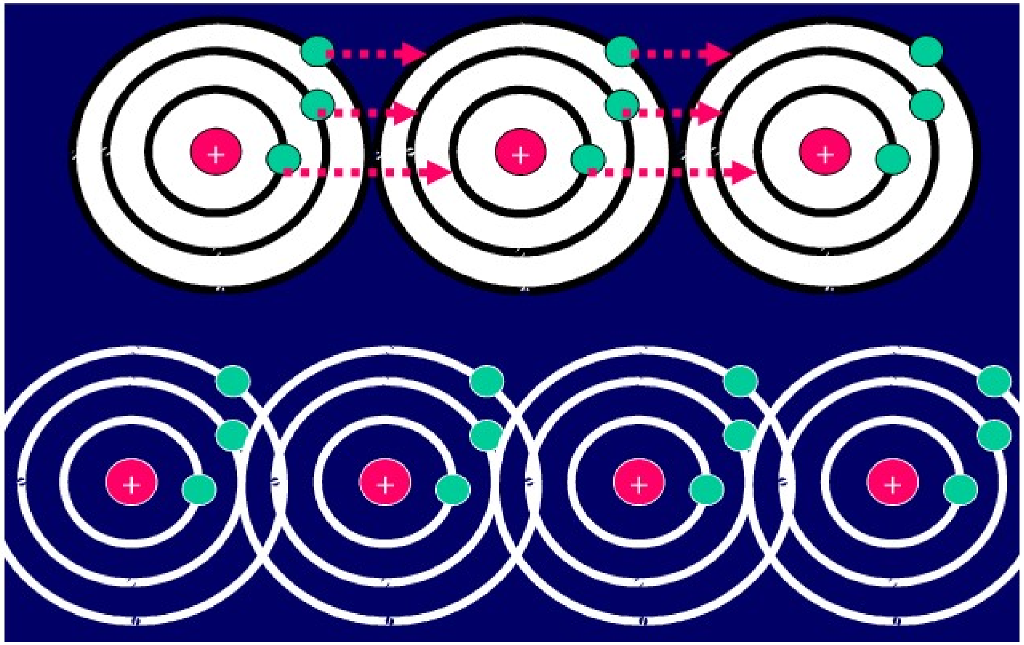


Electrons movement in separated atoms and solids



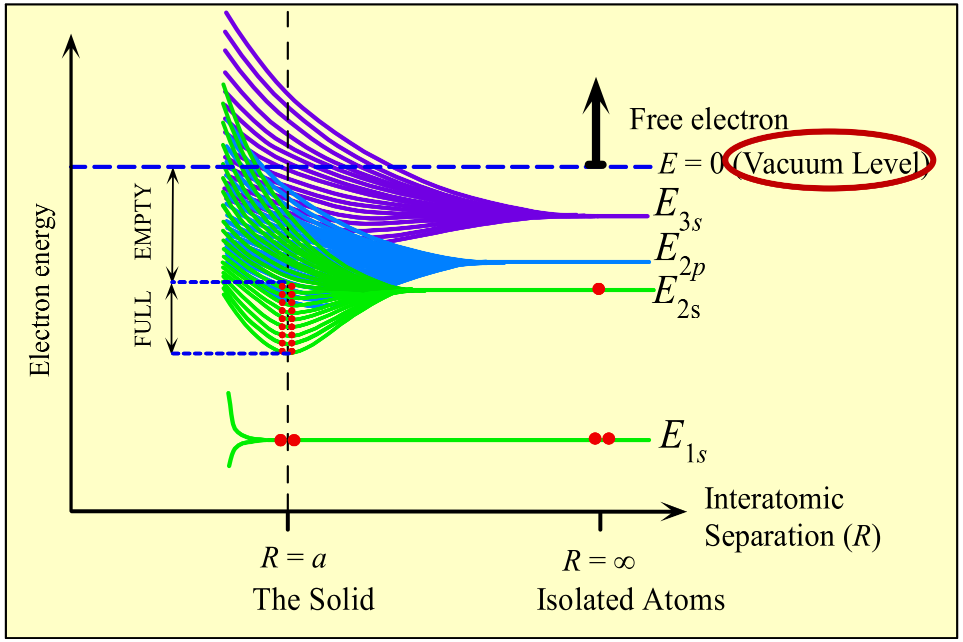
- In the case of separated atoms, electrons move in a fix orbital of individual atom.
- In the case of solid, electrons are shared by all atoms and move in the energy band.

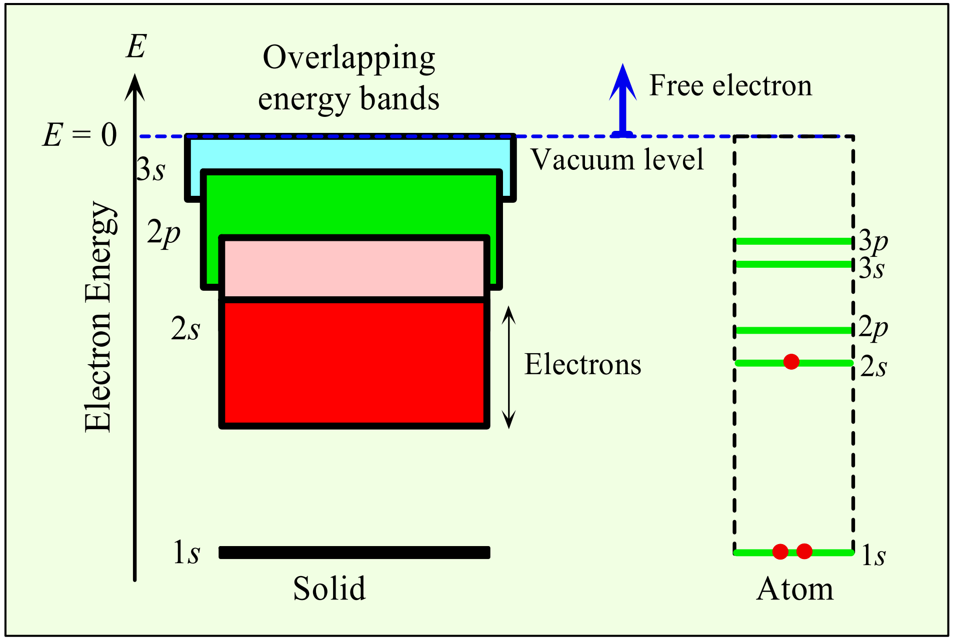
The movement of electrons in crystals



- **Electron Sharing**: The electrons at the atomic level in a crystal are not completely confined in its original orbital, and can be transferred from one to an adjacent atom. As a result, electrons can move throughout the crystal.
- Reason for Electron Sharing: overlap between the electron shells, and the outermost layer of adjacent atoms overlaps.
- Note: Electrons move between similar shells in each atom, and the outer electron shared.

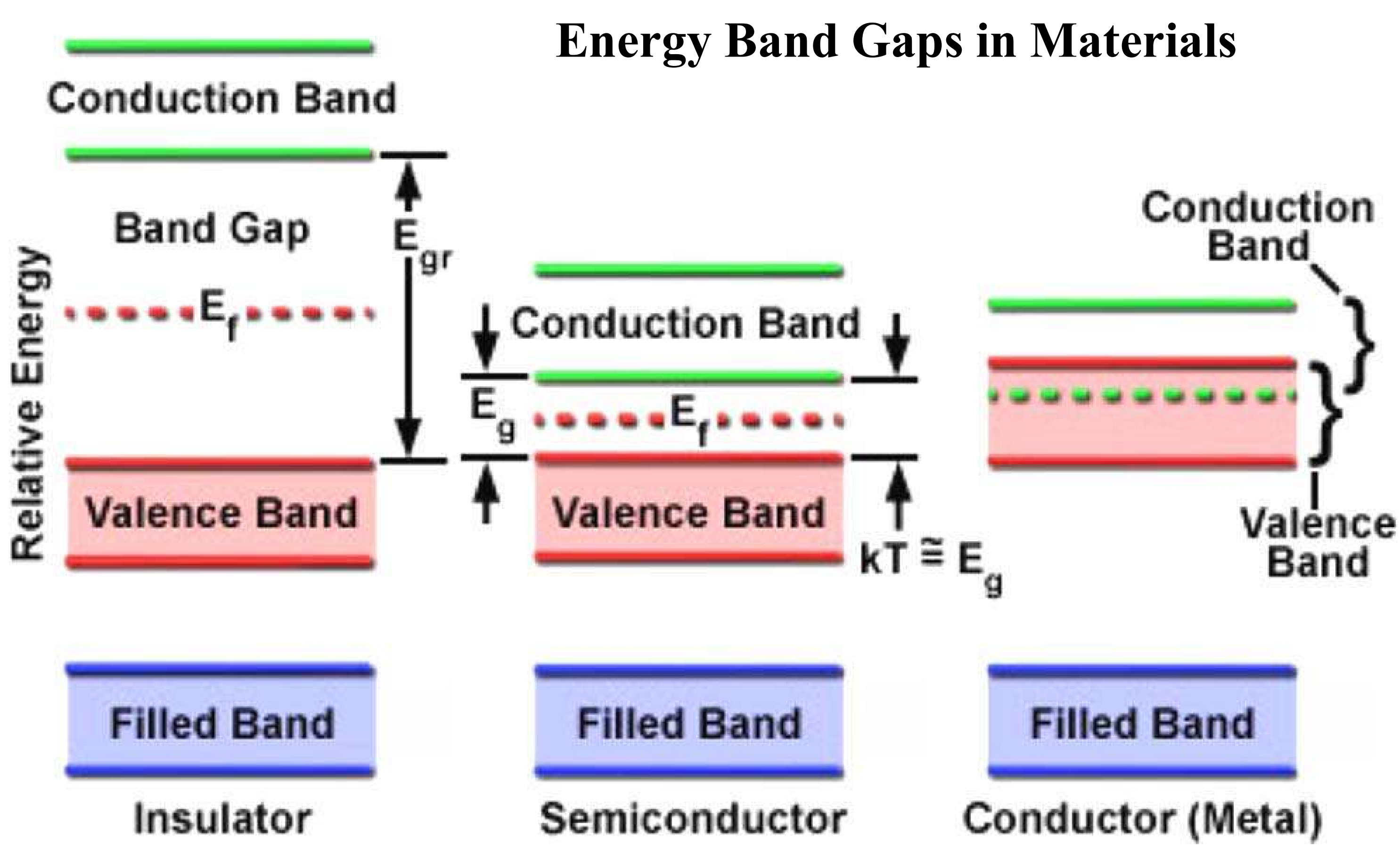
Metal: Bands Overlap





- 2p, 3s and other higher energy level, also spit into finely separated energy levels.
- They overlap with 2s band, hence provide further energy levels and extend the 2s band into higher energy levels.

Insulator, Semiconductor and Metal



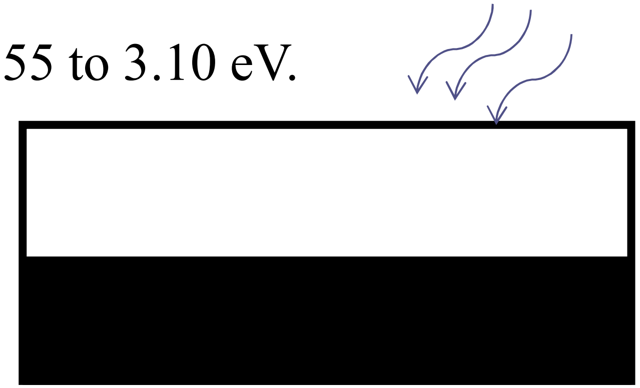
E_g>3 eV E_g<3 eV E_g=0 eV

E_g Vs optical properties of solids

Optical properties of solids are closely related to band structures.

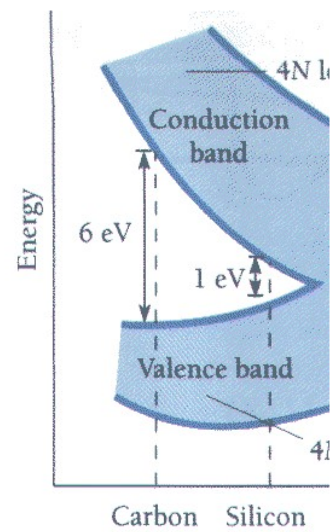
Visible photons have energies ranging from about 1.55 to 3.10 eV.

Metals can absorb visible photons because there are many empty states for electrons to move to.



In diamond, the valence band is full and the conduction band is empty.

A 3 eV photon cannot excite an electron across the 6 eV band gap. Diamond cannot absorb visible photons. Diamonds are transparent.



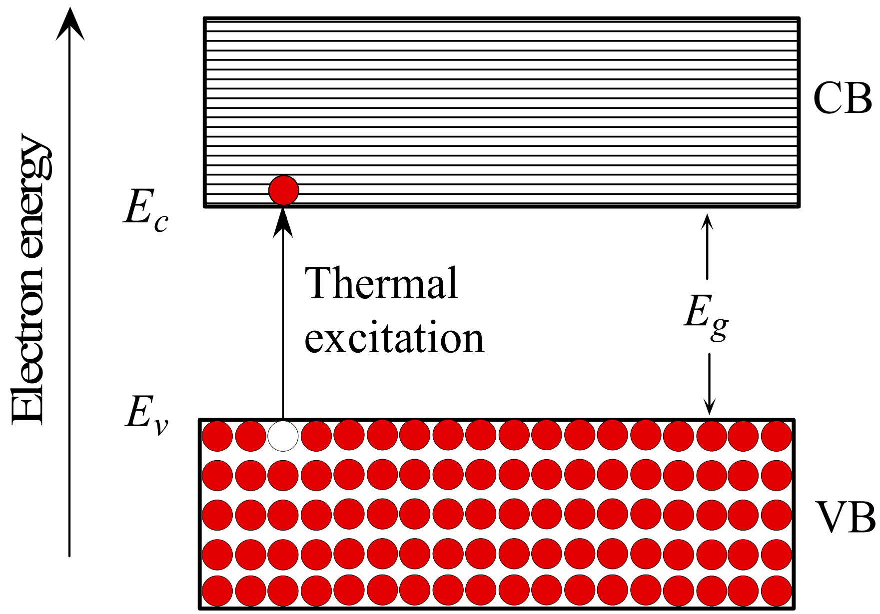
All insulators are transparent?

No. They are opaque because they contain impurities and irregularities which scatter visible light.

Semiconductor

A semiconductor is a material which has a filled (at T=0) valence band separated by a small gap from an empty (at T=0) conduction band.

- Conduction band: can conduct net current, empty band
- Valence band: the highest band contains full *e*.
- Band Gap: energy difference between CB bottom and VB top
- Because the gap is small, at room temperature, there will be a few electrons in the conduction band: semi-conduct



Semiconductor

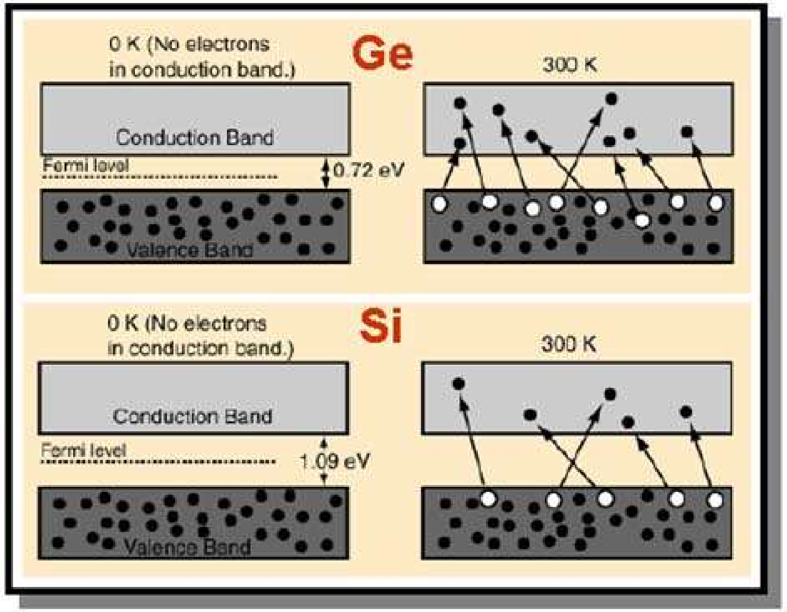
At high temperatures

➔ Some electron excited into conduction band → free electron

➔ Create holes in the valence band → effective free "+" charge

➢ The probability of electron transition across the bandgap is very sensitive to the magnitude of E_g

➢ E_g determines whether a solid is an insulator or a semiconductor



C ~ 6 eV → insulator @ 300 K

(diamond)

Si ~ 1.1 eV

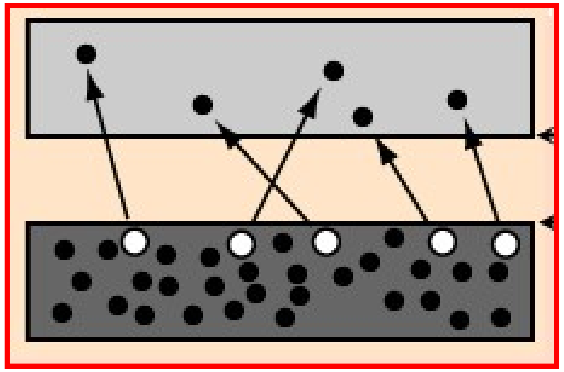
Ge ~ 0.7 eV } semiconductor

T↑ → free electron and holes ↑ → conductivity ↑

Semiconductor

- Metal: electron in the conduction band

- Semiconductor: electrons in the conduction band and holes in the valence band both contribute to current



- In the valence band, electrons can “move into” the empty states, and thus can also contribute to the current

- Hole has positive charge

- Moving direction is opposite to that of electrons

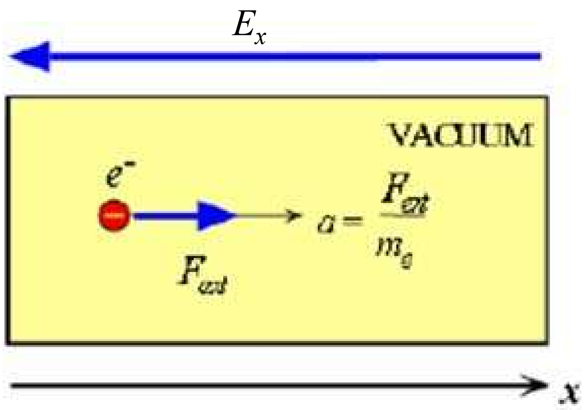
- Hole has larger **effective mass** than that of electron, meaning acceleration *a* is smaller under the same external force, because the movement of holes (represent collective movement of electrons) is difficult than electron.

Electron Effective Mass

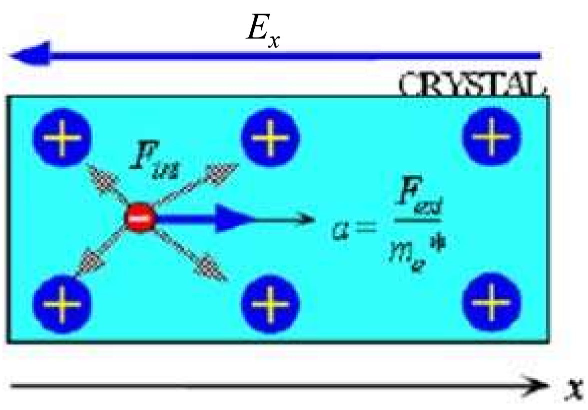
- When an electric field E_x is applied to a metal, an electron near the Fermi level gains energy from the field and moves to higher energy levels
- If the external force $F_{ext} = =eE_x$ is in the x direction, it drives the electron along x

$$\begin{cases} a_{vac} = F_{ext} / m_e \\ a_{cryst} = \dfrac{F_{ext} + F_{\text{int}}}{m_e} \end{cases} \qquad \boxed{a_{cryst} = \dfrac{F_{ext}}{m_e^*}}$$

In QM, instead of the mass in vacuum, we must use the **effective mass** of the 'e' since variation of F_{int} or periodic potential affects on the motion of 'e' in a crystalline solid.



(a) An external force F_{ext} applied to an electron in a vacuum results in an acceleration $a_{vac} = F_{ext} / m_{e*}$



(b) An external force F_{ext} applied to an electron in a crystal results in an acceleration $a_{cry} = F_{ext} / m_{e*}$

Effective Mass

Table 4.2 Effective mass m_e^* of electrons in some metals

Metal	Ag	Au	Bi	Cu	K	Li	Na	Ni	Pt	Zn
$\frac{m_e^*}{m_e}$	0.99	1.10	0.047	1.01	1.12	1.28	1.2	28	13	0.85

Effective Mass	AlAs	GaAs	GaP	InP	InAs	InSb
m_e^* / m_0	0.124	0.067	0.5	0.079	0.024	0.014
m_{hh}^* / m_0	0.5	0.51	0.67	0.65	0.41	0.4

- Larger effective mass means stronger movement resisitance
- Larger effective mass means poor conductivity in general
- Effective mass of hole is larger than that of electron
- Electron mobility is larger than that of hole
  in Si, $\mu_{electron}$=1400 cm^2/Vs, μ_{hole}=450 cm^2/Vs

Q1: In semiconductors, which one has faster mobility, holes or electrons?

$$\sigma = en\mu_d$$

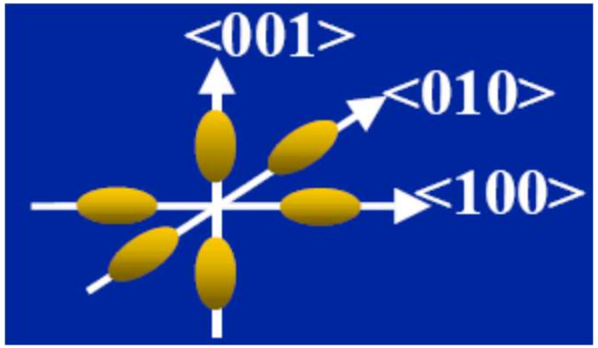
$$\mu_d = \frac{e\tau}{m^*}$$

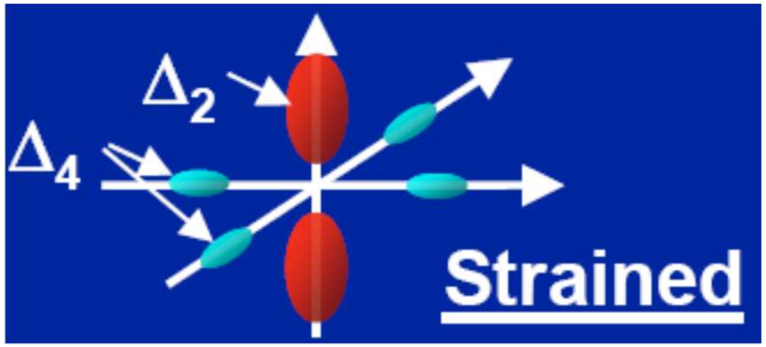
Q2: How to change the effective mass of electron/hole?

Q3: Can the effective mass of electron be negative?

Strain Effect -- Electron

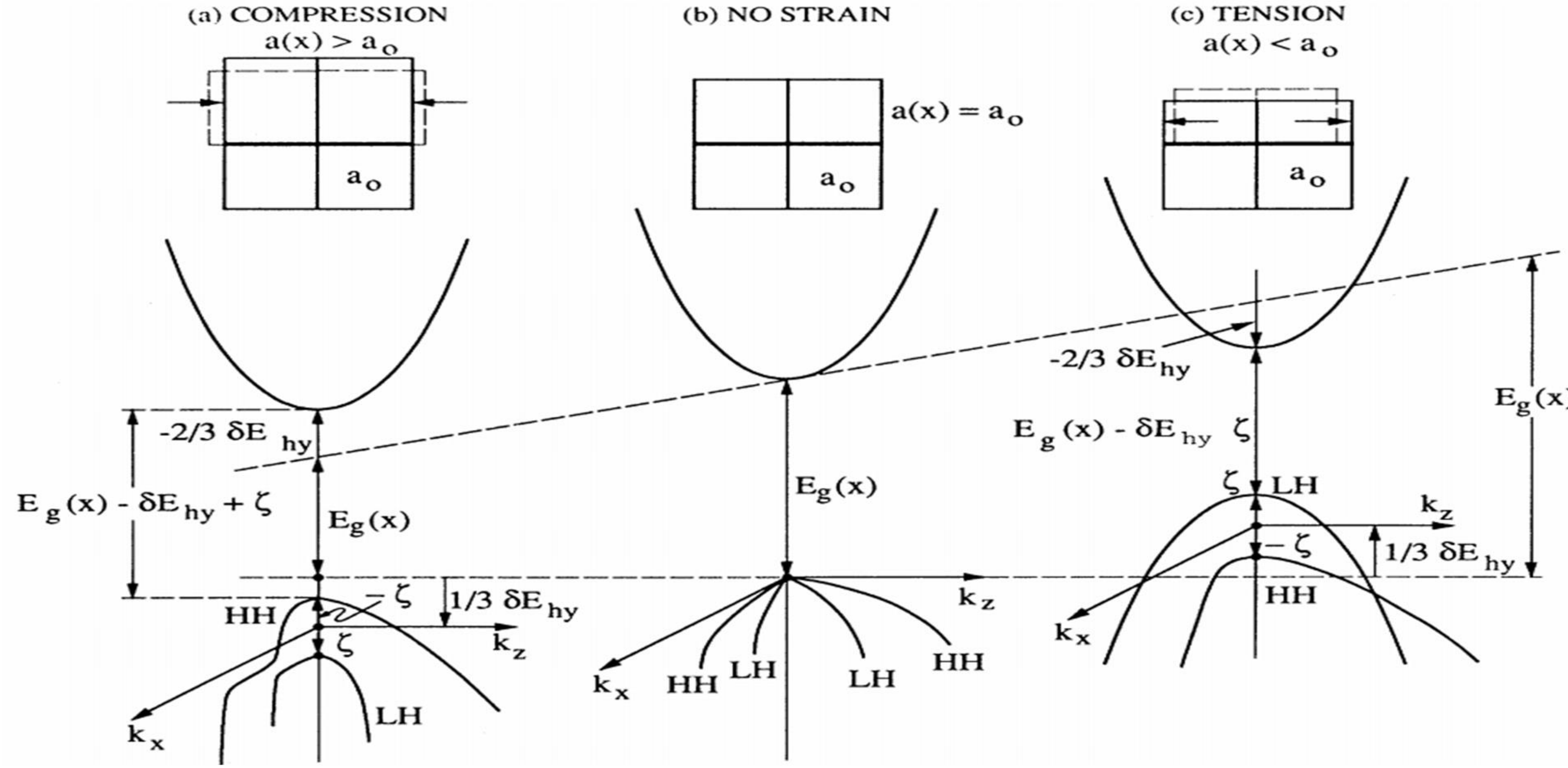
- Electrical symmetry destroyed by strain
  - Four energy valleys go down in energy, two go up (in biaxial strain)
  - Intervally scattering reduced
- Change in curvature
  - Reduction in effective mass



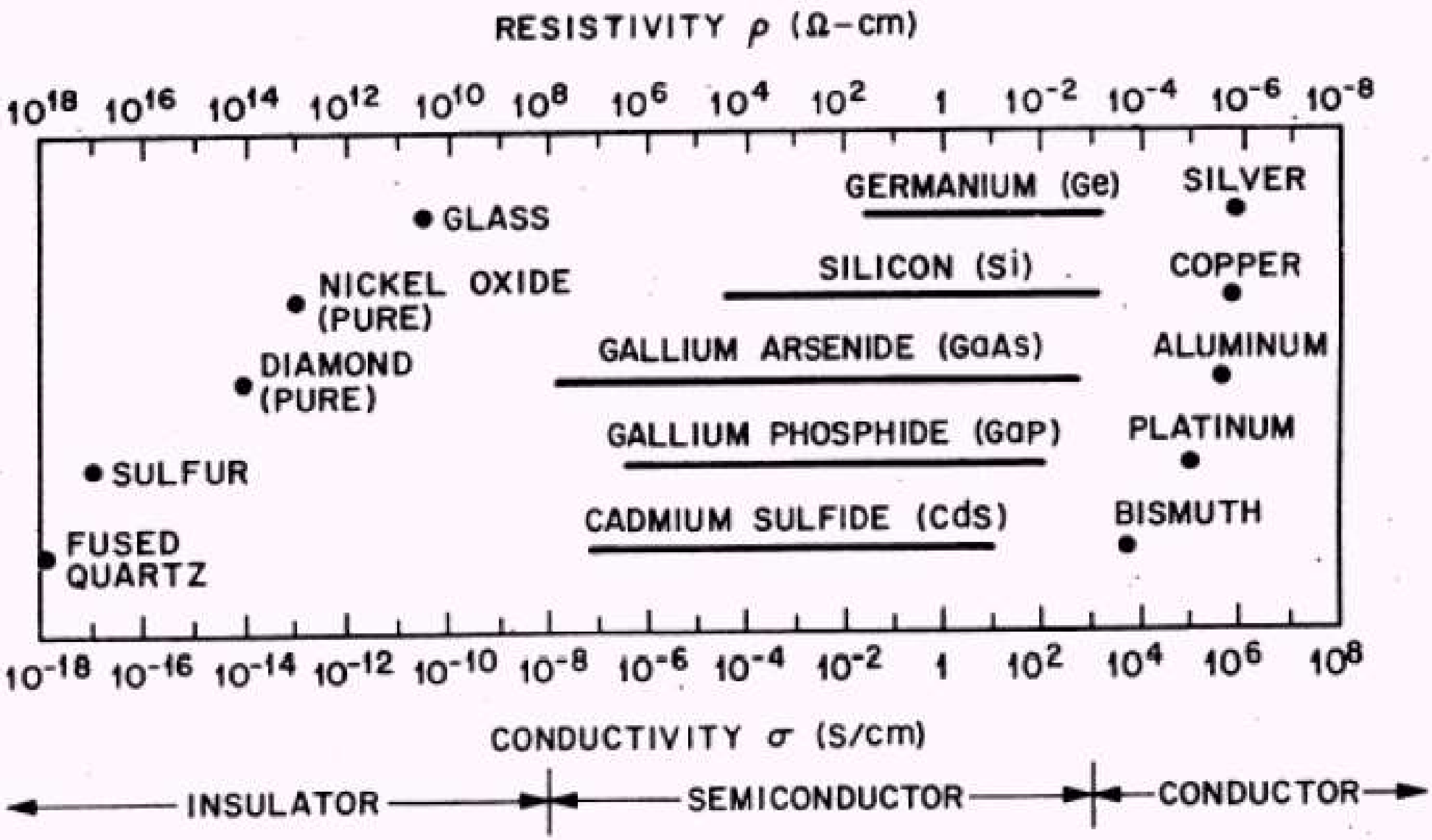


Strain Effect -- Hole

- Unaxial strain reduces effective mass
- Biaxial strain splits LH and HH bands, reduces scattering



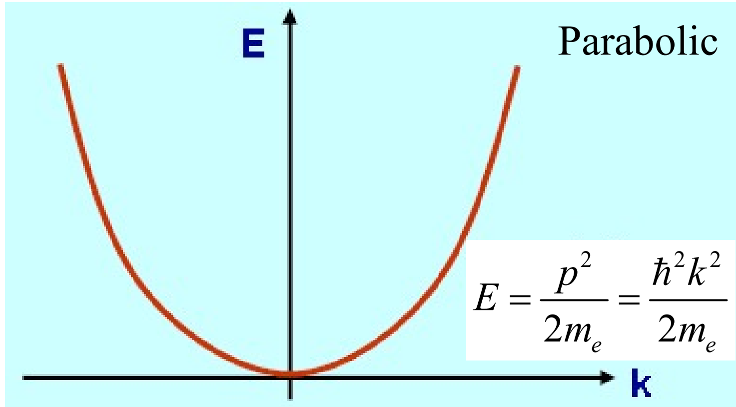
Insulator, Semiconductor and Metal



Typical range of conductivities for insulators, semiconductors, and conductors.

Advanced Band Theory

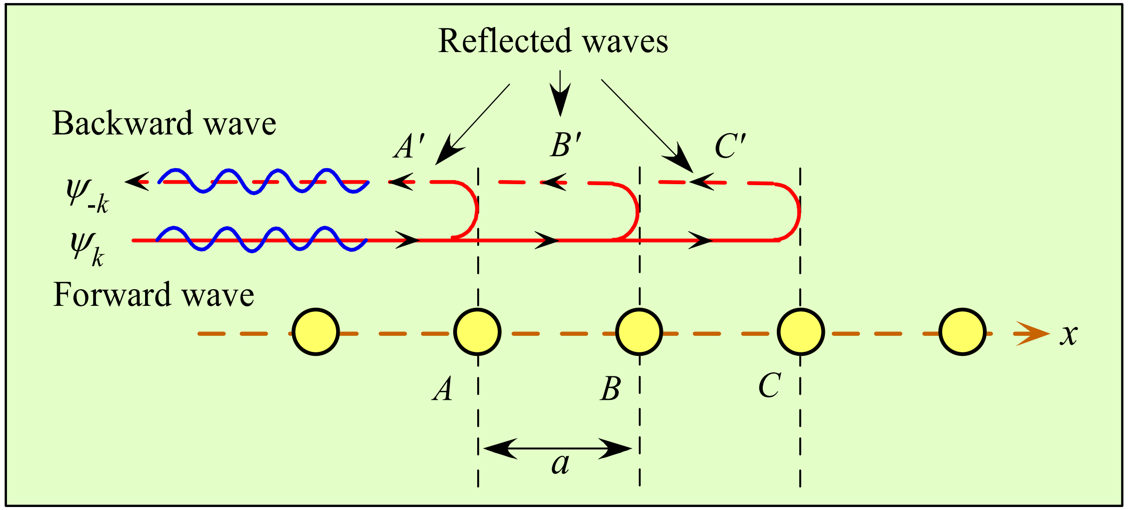
- For free electron, travelling wave $\varphi(k) = A\exp(jkx)$ the E-k relationship is



- For electron move in a crystal, there is reflective wave and forward wave.
- Reflective waves interfere with each other constructive if

$$2a = n\lambda = n\frac{2\pi}{k} \Rightarrow k = \frac{n\pi}{a}$$

- Combination of forward wave and reflective wave results in standing wave which does not propagate



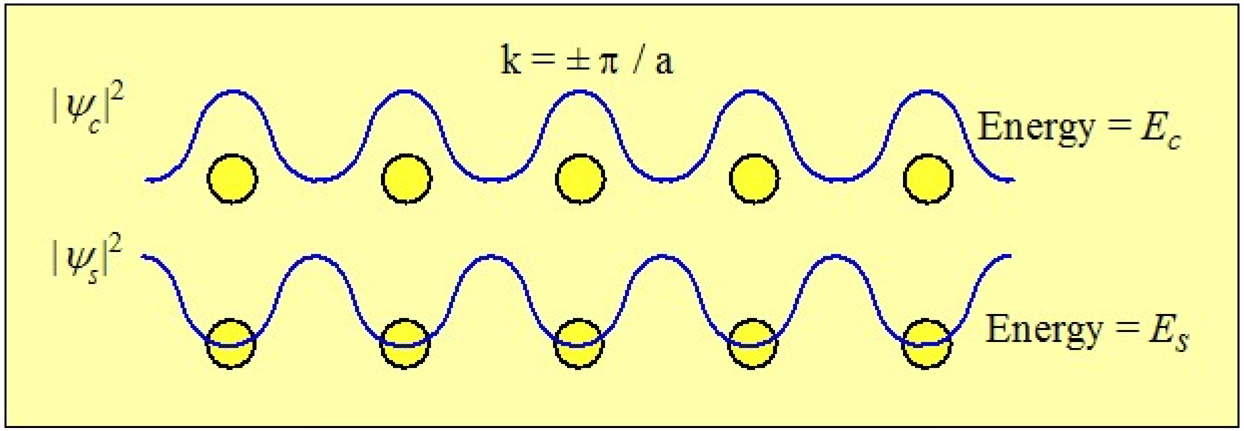
An electron wave propagation through a linear lattice. For certain k values the reflected waves at successive atomic planes reinforce each other to give rice to a reflected wave travelling in the backward direction. The electron then cannot propagate through the crystal

Advanced Band Theory

- Reflective wave: $\varphi_{-k}(x) = A\exp(-jkx)$, only exists for $k = \dfrac{n\pi}{a}$, $n = 1,2,3$

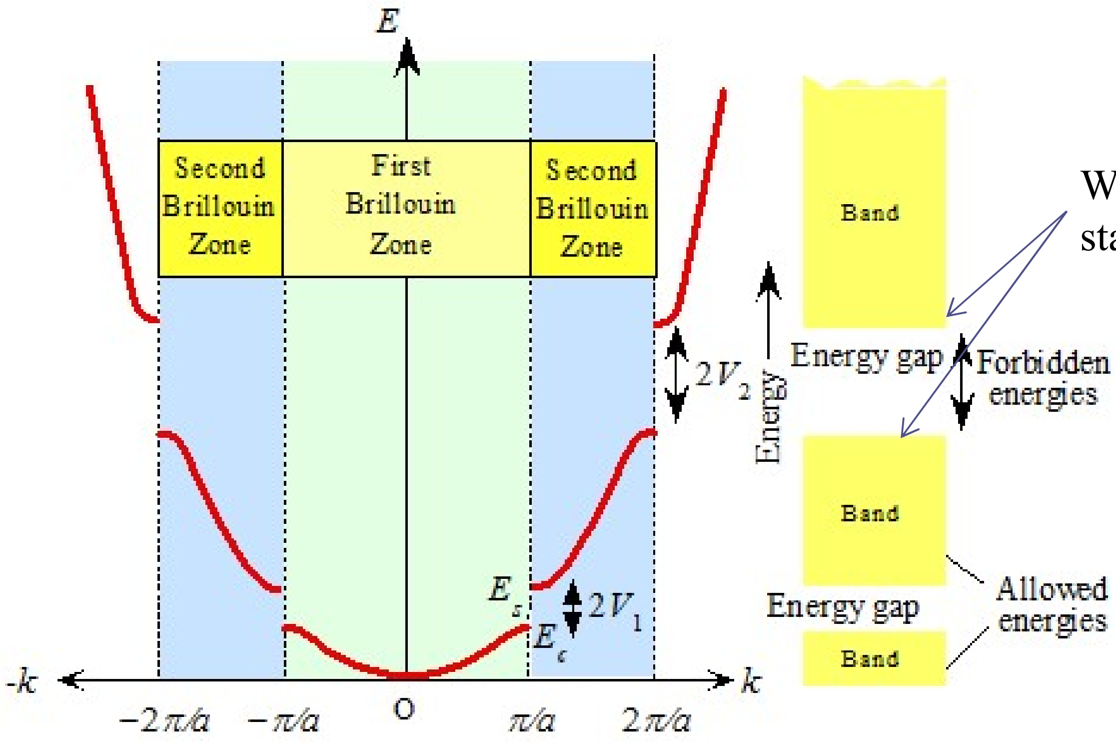
- Forward wave: $\varphi_k(x) = A\exp(jkx)$

- Two possible standing waves: $\begin{cases} \varphi_c(x) = A\exp(jkx) + A\exp(-jkx) = A_c\cos\left(\dfrac{n\pi x}{a}\right) \\ \varphi_s(x) = A\exp(jkx) - A\exp(-jkx) = A_s\sin\left(\dfrac{n\pi x}{a}\right) \end{cases}$



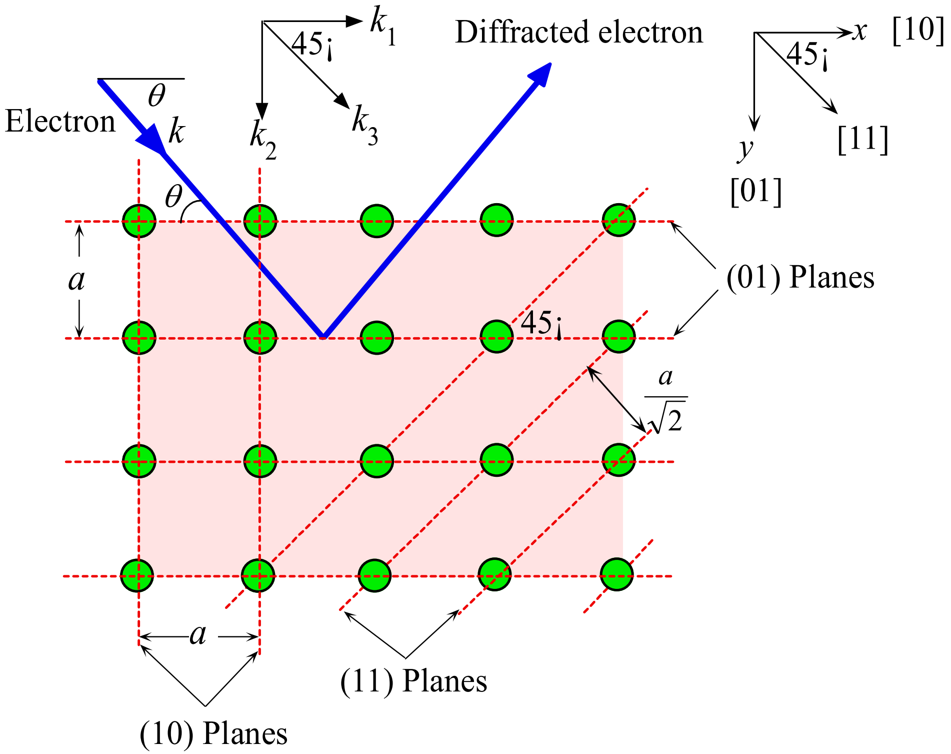
Advanced Band Theory

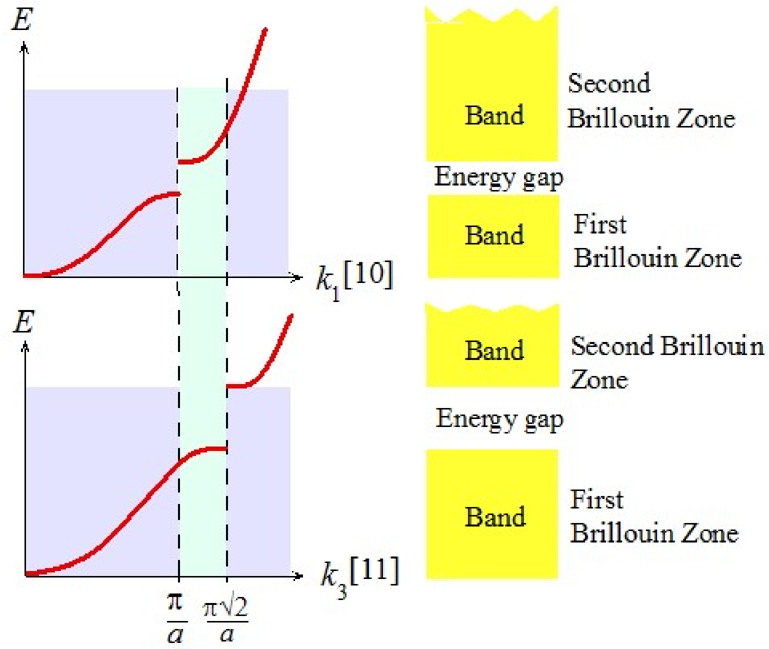
- Potential energy for $\varphi_c(x)$ is: $V_c = \frac{1}{L}\int_o^L V(x)|\varphi_c(x)|^2 = -V_n$
- Potential energy for $\varphi_s(x)$ is: $V_s = \frac{1}{L}\int_o^L V(x)|\varphi_c(x)|^2 = V_n$
- The energy for two standing wave is $E_c = \frac{(\hbar k)^2}{2m_e} - V_n, \quad E_s = \frac{(\hbar k)^2}{2m_e} + V_n, \quad k = \frac{n\pi}{a}$



Wave cannot propagate standing wave

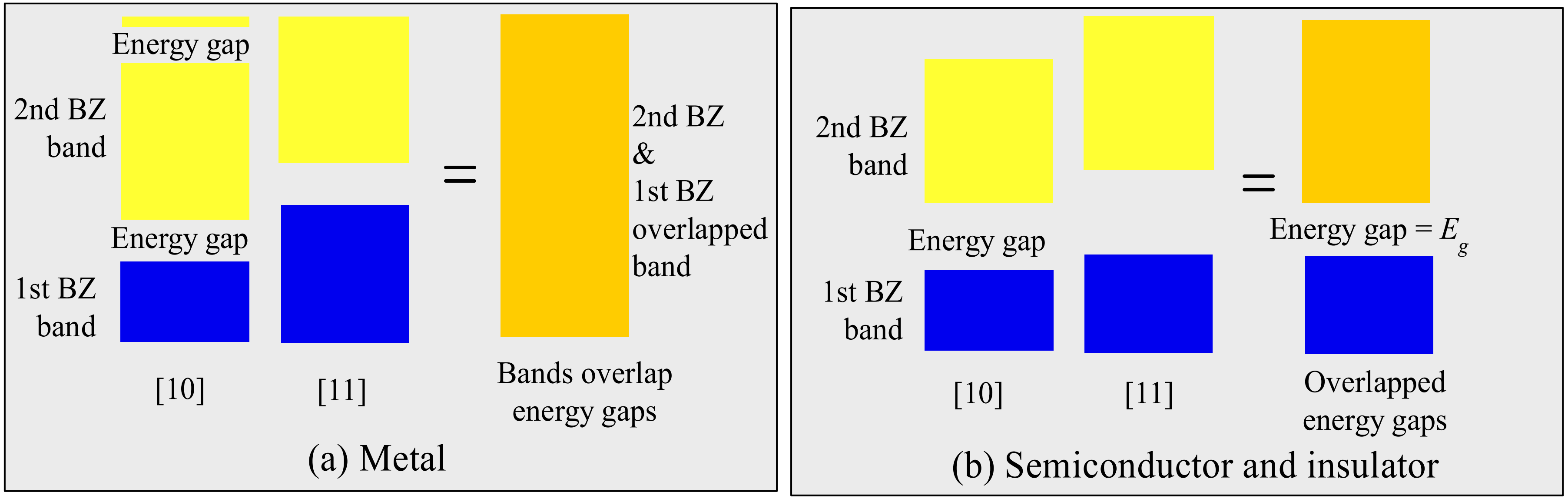
In the Case of 3D Diffraction





- Reflection wave interfere constructive when $k = \dfrac{n\pi}{a}$
- In [10] and [01] plane, $d = a \Rightarrow k = \dfrac{n\pi}{a}$
- In [11] plane, $d = \dfrac{a}{\sqrt{2}} \Rightarrow k = \dfrac{n\pi\sqrt{2}}{a}$

Metal, Semiconductor and Insulator



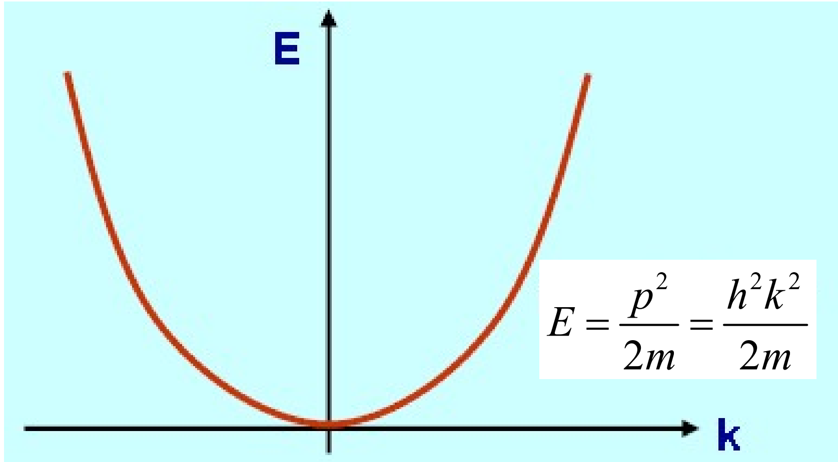
(a) Metal: For the electron in a metal there is no apparent energy gap because the 2^{nd} BZ (Brillouin Zone) along [10] overlaps the 1^{st} BZ along [11]. Band overlap the energy gaps. Thus the electron can always find any energy by changing its direction.

(b) Semiconductor or insulator: For the electron in a semiconductor there is an energy gap arising from the overlap of the energy gaps along [10] and [11] directions. The electron can never have an energy within this energy gap, E_g.

E-*k* Diagram: Velocity and Effective Mass

$$E_c = \frac{p^2}{2m} = \frac{(\hbar k)^2}{2m}$$

$$\frac{dE}{dk} = \frac{\hbar^2 k}{m} = \frac{\hbar p}{m} = \hbar v \Rightarrow v = \frac{1}{\hbar}\frac{dE}{dk}$$



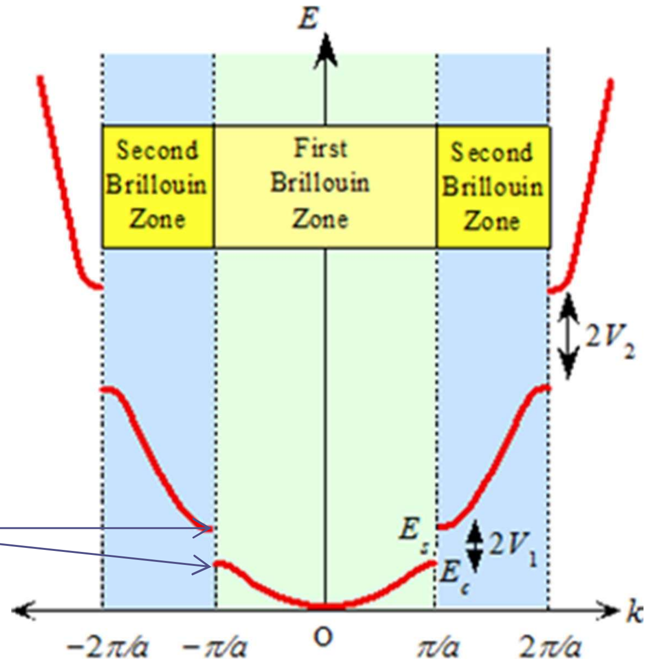
- Velocity is proportional to the slope of the *E*-*k* curve

$$\frac{d^2E}{dk^2} = \frac{\hbar^2}{m} \Rightarrow \frac{1}{m} = \frac{1}{\hbar^2}\frac{d^2E}{dk^2}$$

- Mass is reverse proportional to the curvature of the *E*-*k* curve

Slope approach 0 → Velocity approach 0

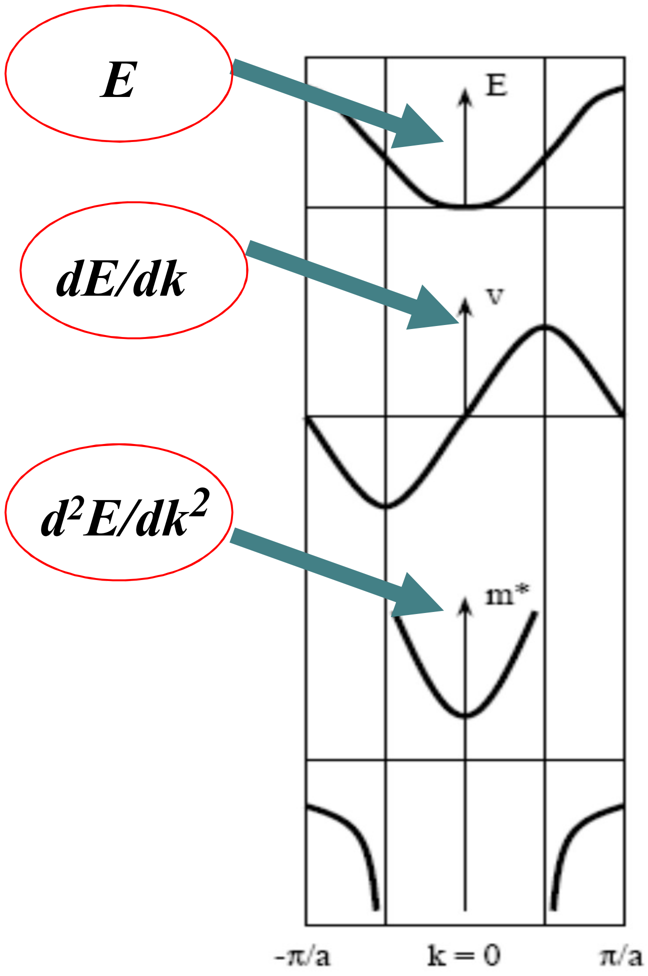
Curvature is small → effective mass is larger



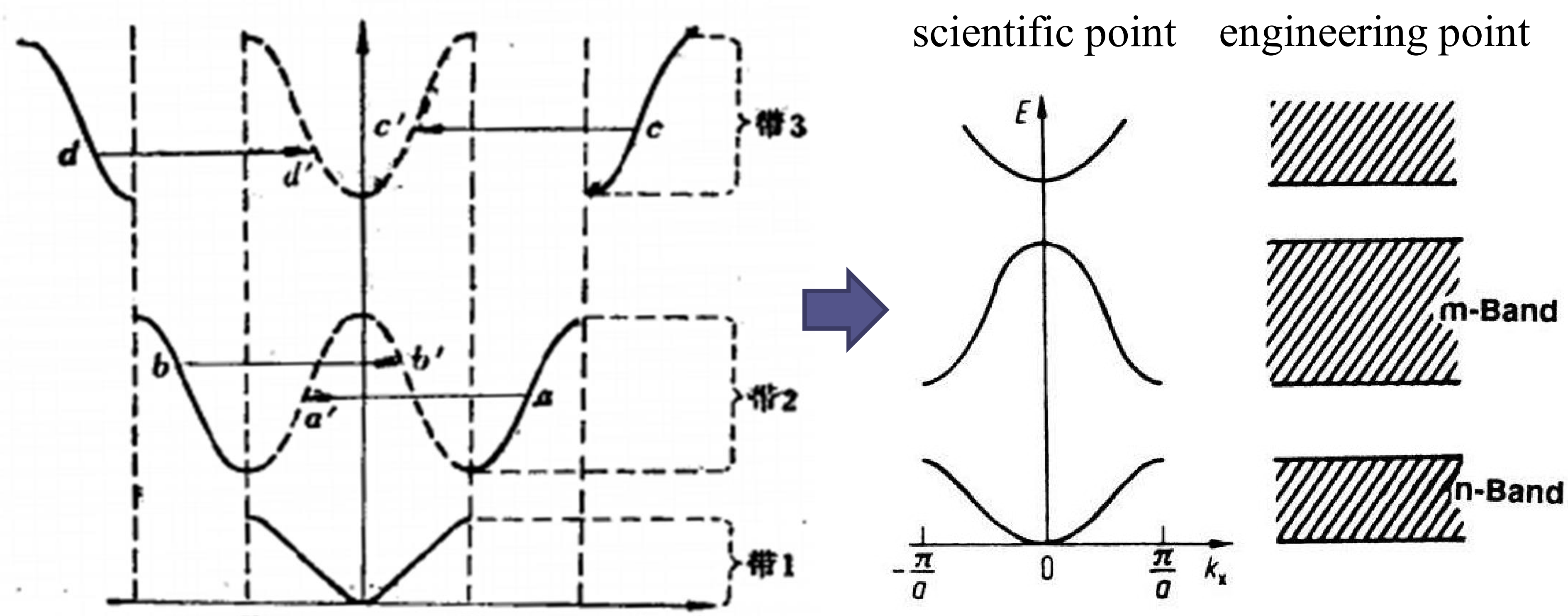
E-k Diagram: Velocity and Effective Mass

At k=0, electron has a constant positive value and it rises rapidly as k value increases. After experiencing a singularity (infinite mass) the effective mass becomes negative up to the top

- Therefore:
- - *m** is positive near the bottoms of all bands,
- - *m** is negative near the tops of all bands.



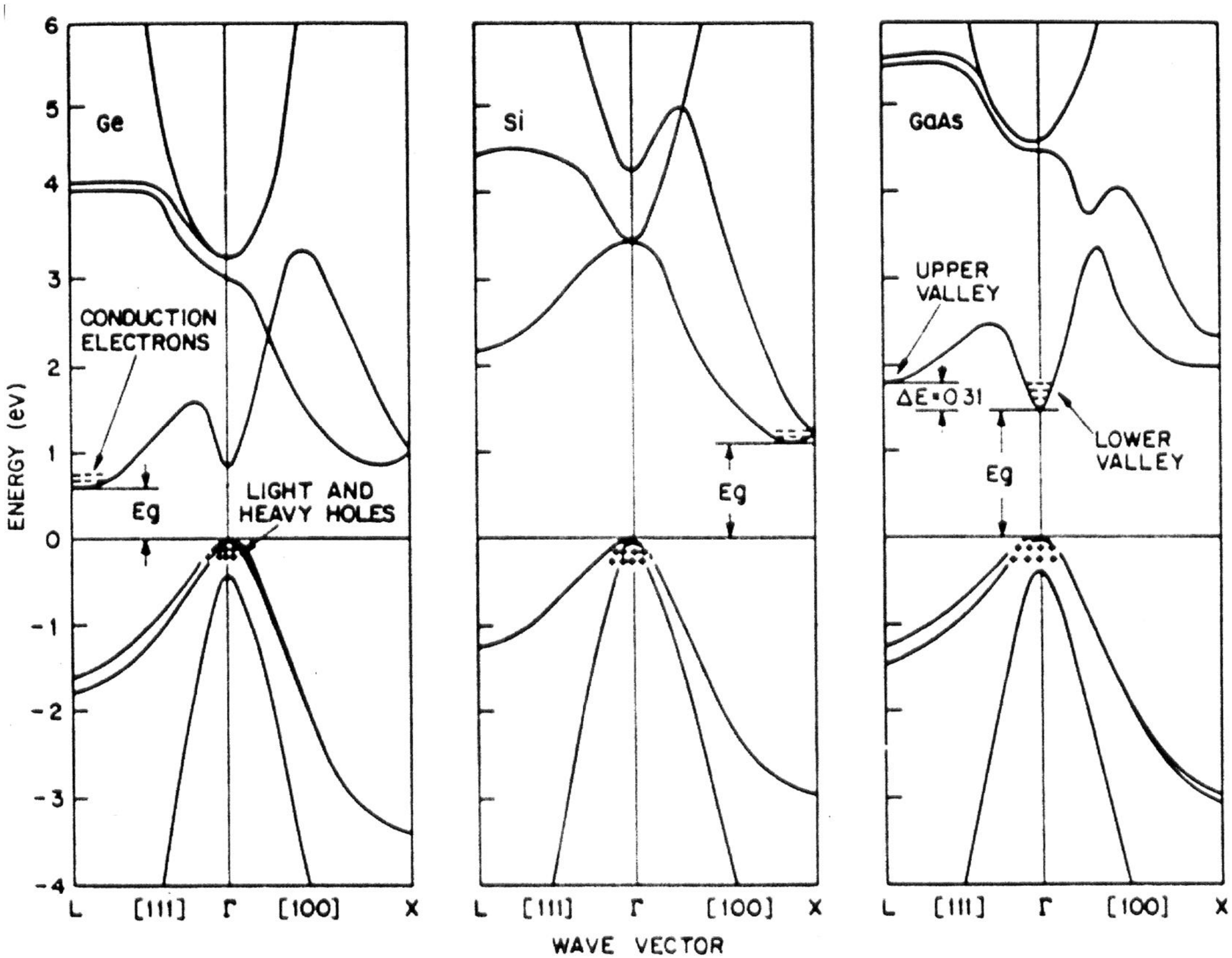
Reduced Brillouin Zone （简约布里渊区）



Reduced Brillouin Zone

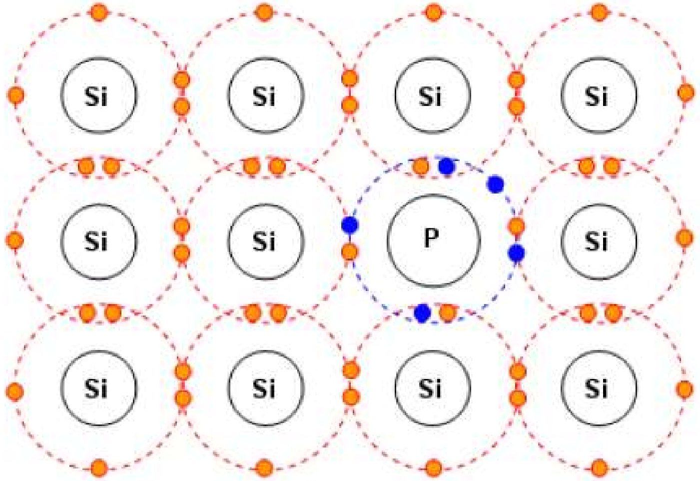
Wave vector k=2π/a is identical to the wave vector k=0. The second zone boundary occurs at k=2π/a which is the same as saying that the second zone boundary occurs at k=0.

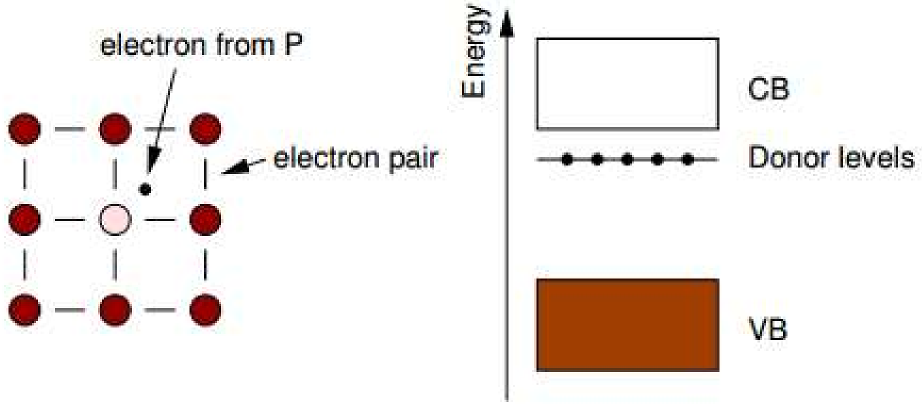
E-k Diagram for Real Semiconductors



Doping in Semiconductor (*n*-doped)

Can alter the conductivity of semiconductor by introducing excess carriers $\sigma = en\mu_d$





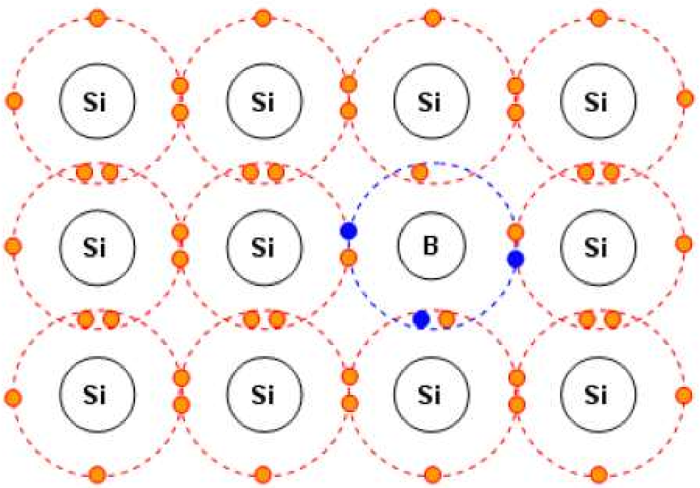
As compared to Si, the Phosphorus has one extra valence electron which, after all bounds are made, has very weak bounding.

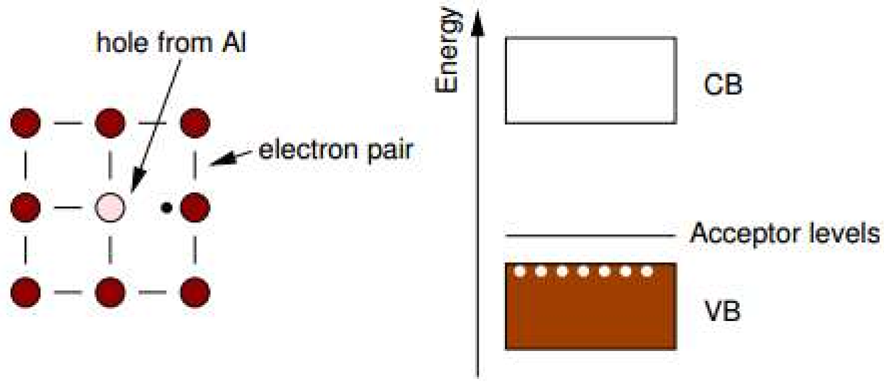
Phosphorus create donor levels in the band gap → electrons can jump from the donor levels to the CB easily (even under RT) → create excess electrons → increase conductivity

This type of impurity is called donor

Doping in Semiconductor (*p*-doped)

Can alter the conductivity of semiconductor by introducing excess carriers $\sigma = qn\mu_d$





Boron has only three valence electrons, one electron less than the Si atom.

Having only three valence electrons → not enough to fill all four bonds → it creates an excess hole that can be used in conduction.

This type of impurity is called acceptor

Doping in Semiconductors

Ideal Semiconductor

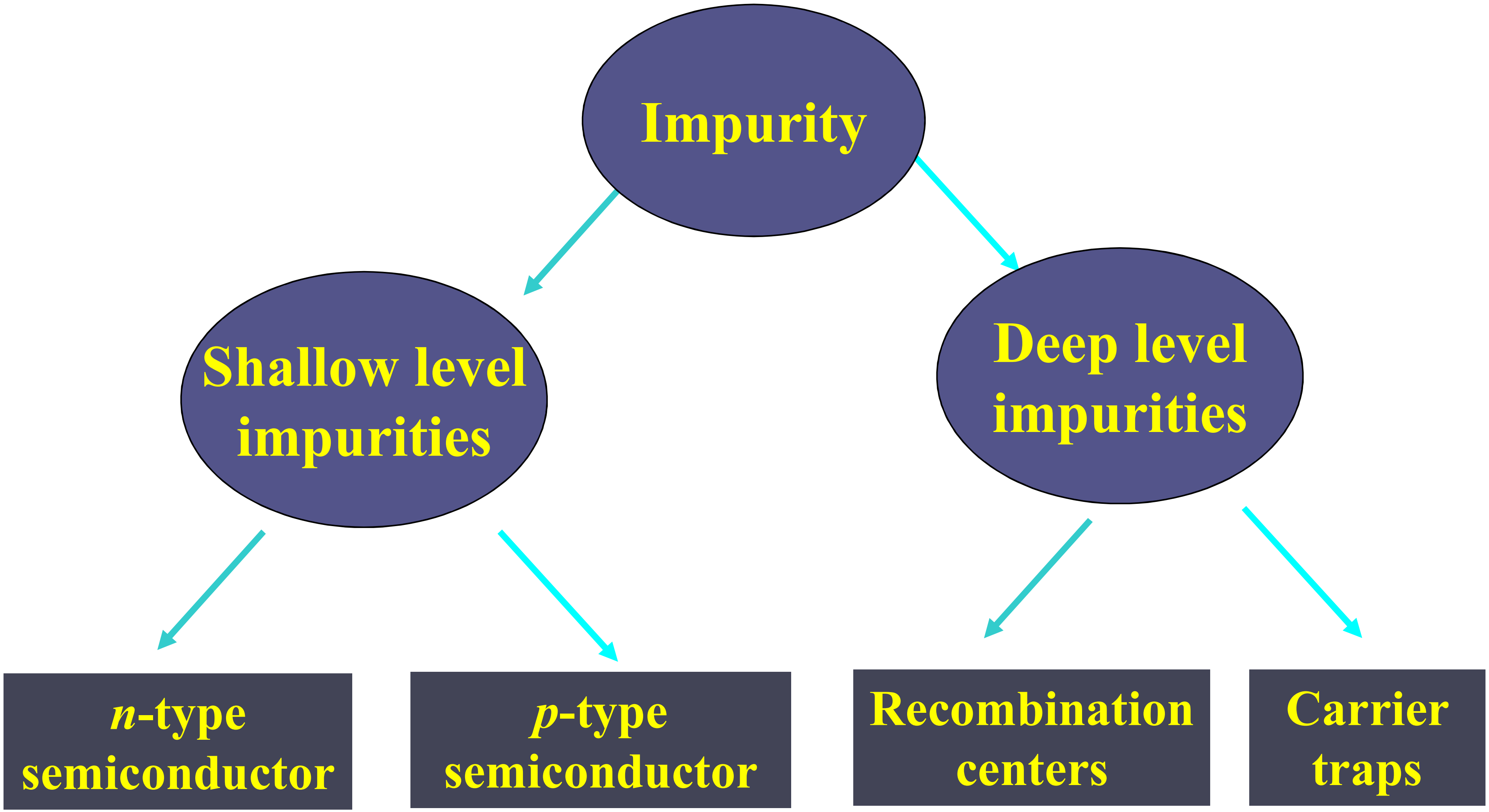
- The atoms are strictly periodic and have a complete lattice structure.
- There are no impurities in the crystal and no defects.
- The electrons perform a common motion in the periodic field, forming a band and a forbidden band—the electron energy can only be at the energy level in the band, and no energy level inside the bandgap.

⇨ **Intrinsic Semiconductor:** The crystal has a complete (perfect) lattice structure without any impurities or defects. Carriers are provided by intrinsic excitation.

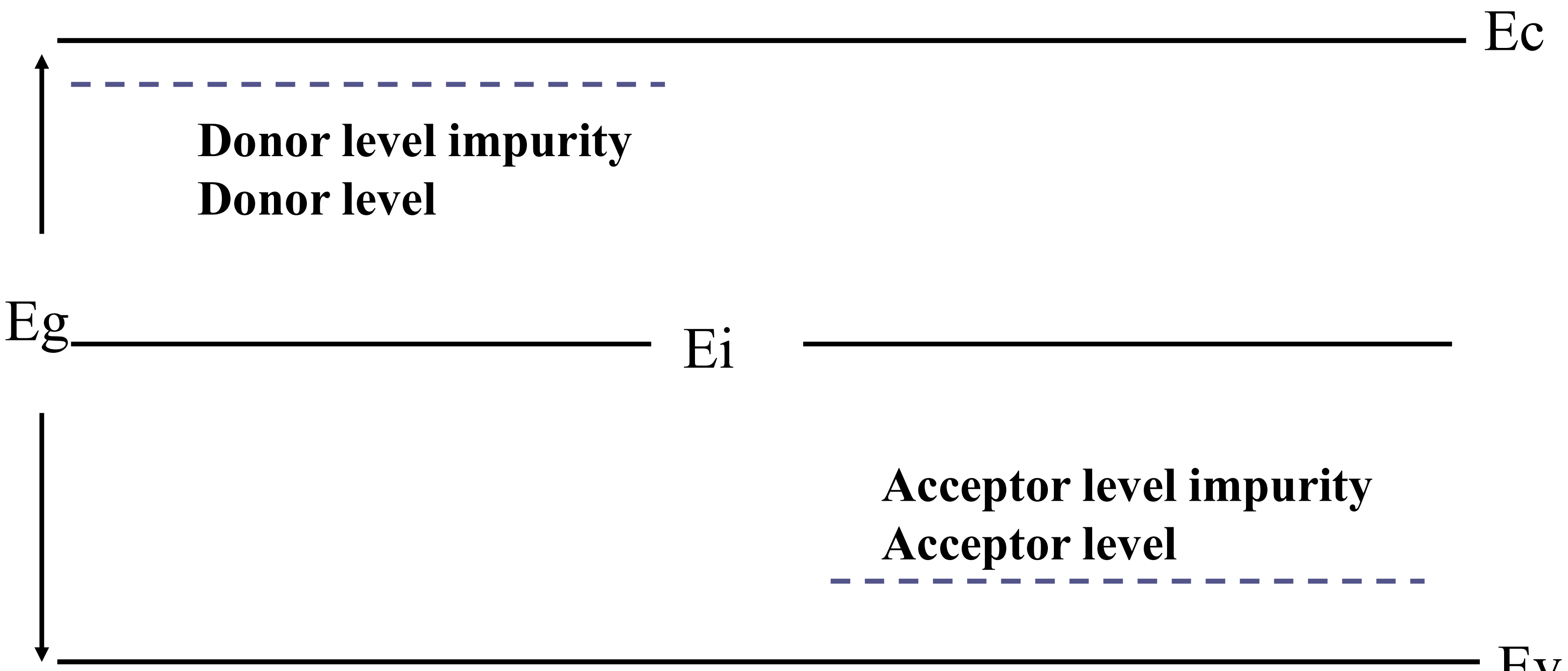
Real Semiconductor

- In an actual semiconductor, atoms are not stationary at a lattice position with strict periodicity, but vibrate near their equilibrium position;
- The actual semiconductor is not pure, and contains impurities;
- The actual semiconductor lattice structure is not intact, but there are various forms of defects, point defects, line defects, surface defects;

⇨ Impurities and defects can introduce energy levels in the forbidden band, which has a decisive effect on the properties of semiconductors.

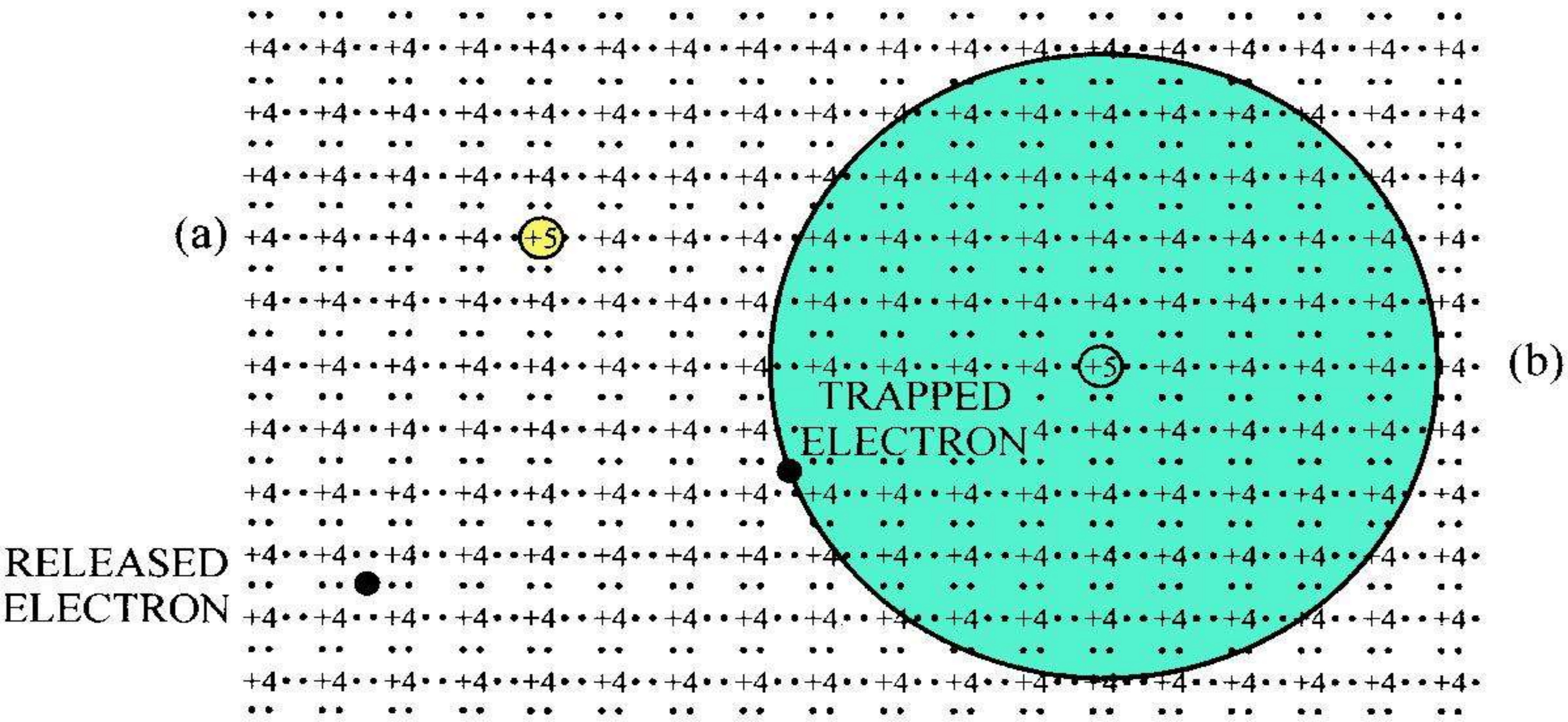

Impurity
Shallow level impurities
Deep level impurities
n-type semiconductor
p-type semiconductor
Recombination centers
Carrier traps

Energy levels



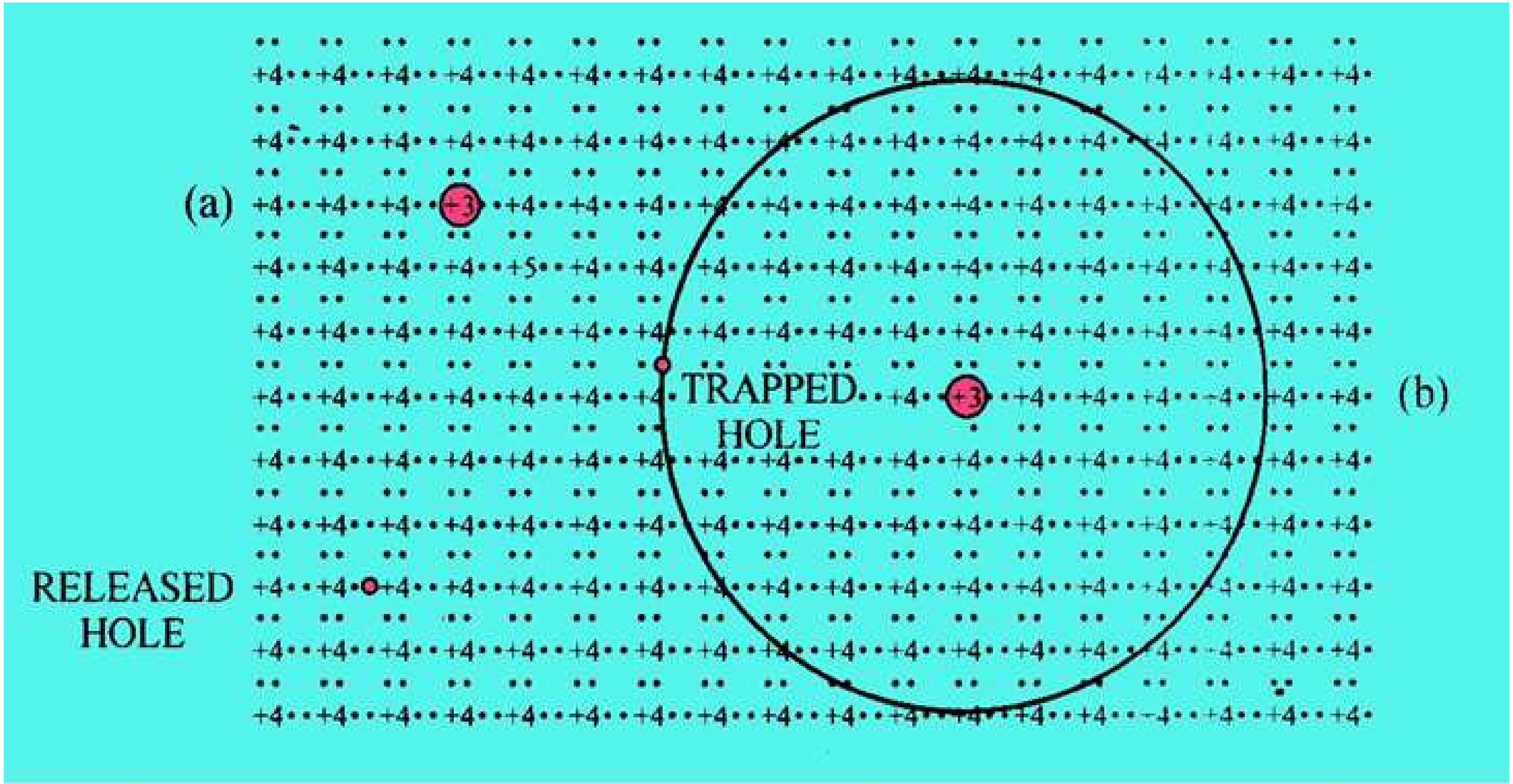
Donor

(a) Ionized state　　　(b) Neutral donor level

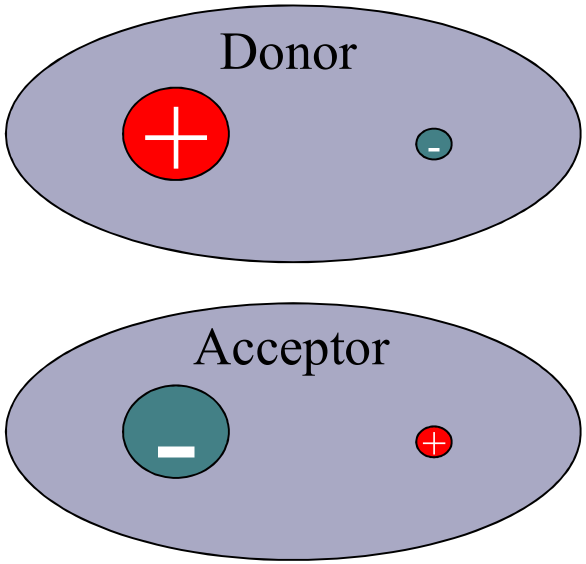


Acceptor

(a) Ionized state (b) Neutral acceptor level



Ionization energy of shallow level impurities



Hydrogen-like model

Shallow level impurity = impurity ion + bound electron (hole)

Calculate the orbital radius and ionization energy of bound electrons or holes

Radius of rotation: $r = \dfrac{\varepsilon_r \varepsilon_o h^2}{m^* \pi q^2} n^2$

Ionized Energy: $\Delta E = \dfrac{m^* q^4}{8\varepsilon_r{}^2 \varepsilon_o{}^2 h^2} = \dfrac{m^*}{m_o} \dfrac{1}{\varepsilon_r{}^2} \dfrac{m_o q^4}{8\varepsilon_o{}^2 h^2} = \dfrac{m^*}{m_o} \dfrac{1}{\varepsilon_r{}^2} E_0$

Ionization energy of donor impurity

$$\Delta E_D = \frac{m_n^* q^4}{8\varepsilon_r^2 \varepsilon_0^2 h^2} = \frac{m_n^*}{m_0} \frac{E_0}{\varepsilon_r^2}$$

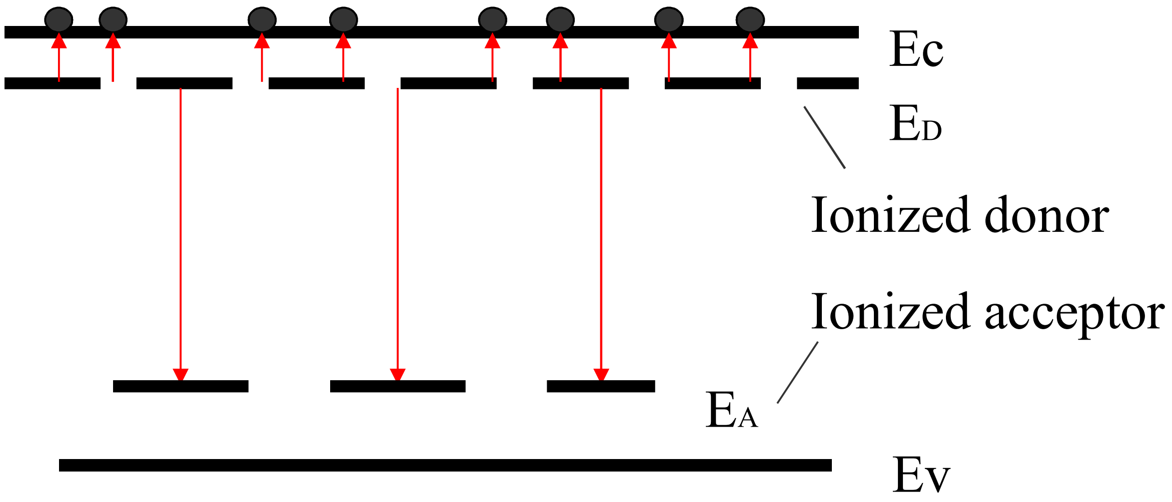
Ionization energy of acceptor impurity

$$\Delta E_A = \frac{m_p^* q^4}{8\varepsilon_r^2 \varepsilon_0^2 h^2} = \frac{m_p^*}{m_0} \frac{E_0}{\varepsilon_r^2}$$

Impurity Compensation

Both donor and acceptor impurities exist in the semiconductor, and the donor and the acceptor cancel each other

(1) $N_D > N_A$



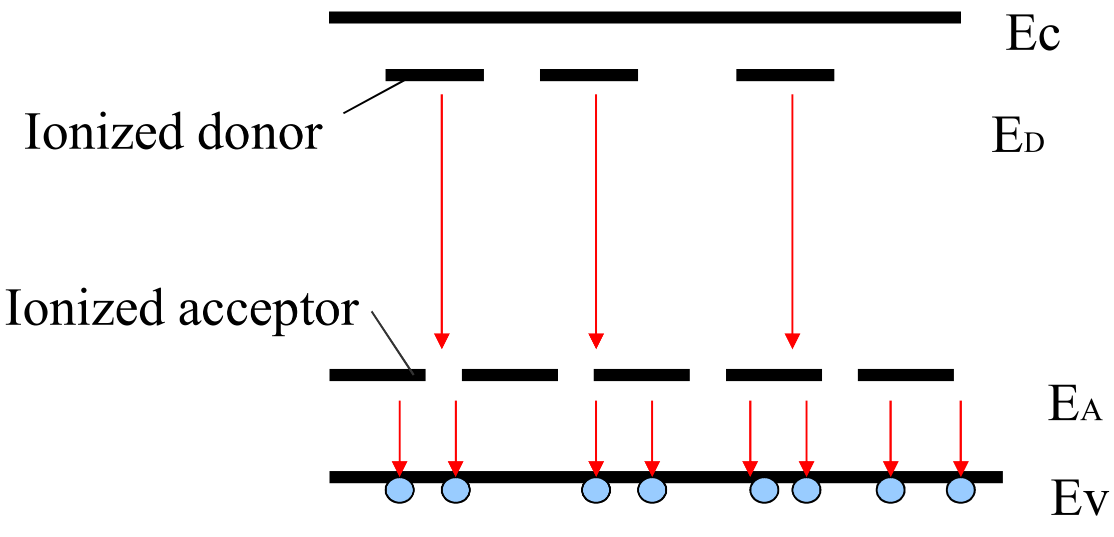
Effective donor concentration $n = N_D - N_A$

At this time, the semiconductor is an ***n*-type semiconductor**

Impurity Compensation

Both donor and acceptor impurities exist in the semiconductor, and the donor and the acceptor cancel each other

(2) $N_D < N_A$



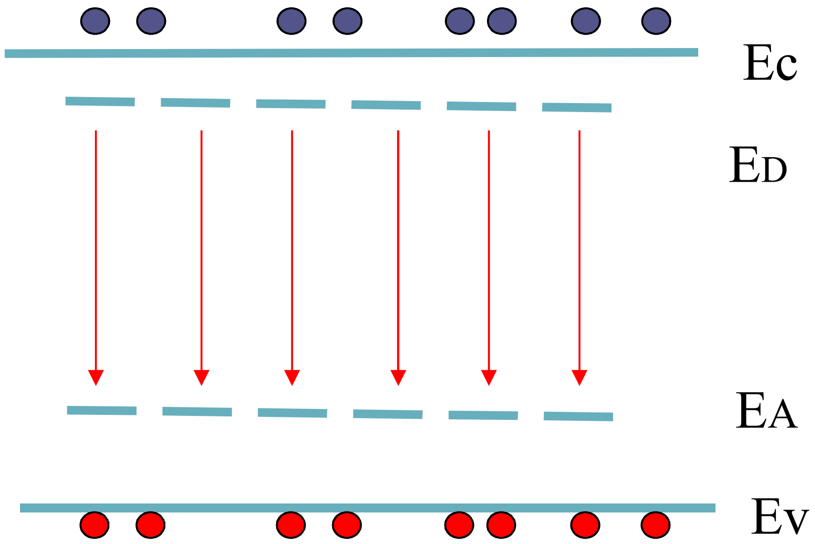
Effective donor concentration $n = N_A - N_D$

At this time, the semiconductor is an ***p*-type semiconductor**

Impurity Compensation

Both donor and acceptor impurities exist in the semiconductor, and the donor and the acceptor cancel each other

(3) $N_D \approx N_A$



Conduction band electrons generated by intrinsic excitation

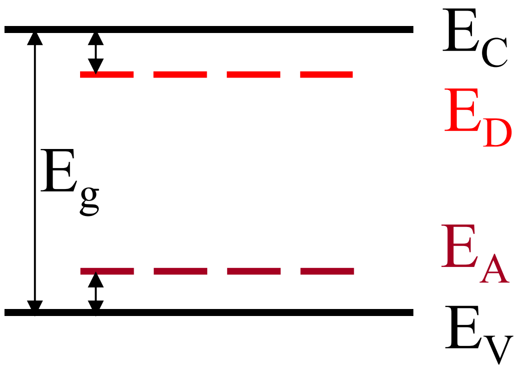
Valence band holes generated by intrinsic excitation

High compensation of impurities

Deep Impurity Level

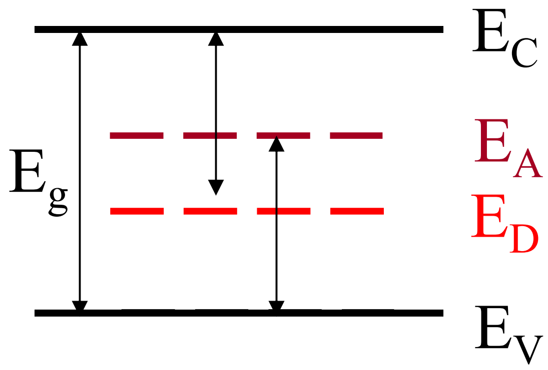
According to the position of the impurity level in the forbidden band, the impurities are divided into:

The shallow level impurity → energy level is close to the bottom of the conduction band E_C or the top of the valence band E_V, the ionization energy is very small



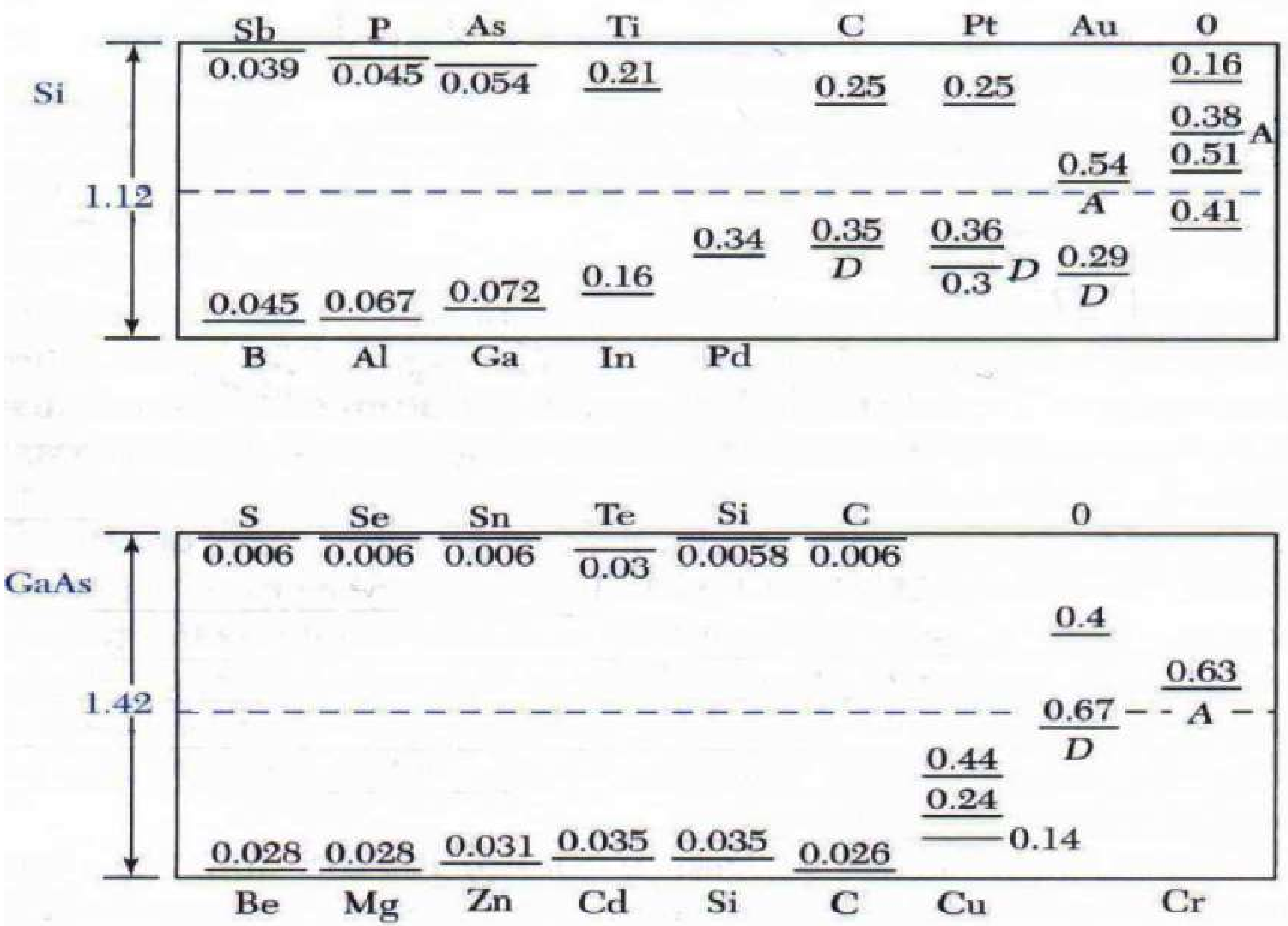
$$\Delta E_D << E_g \qquad \Delta E_A << E_g$$

Deep level impurities → energy level away from the bottom of the conduction band E_C or the top of the valence band E_V, the ionization energy is larger



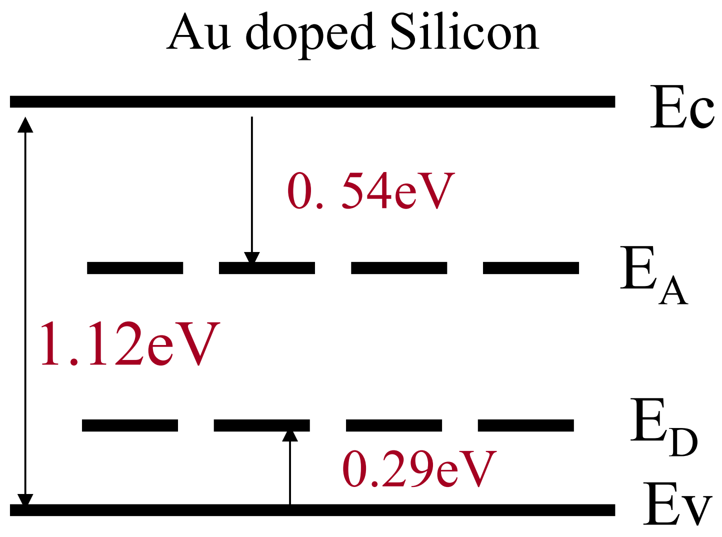
Deep Impurity Level

Deep and shallow impurities levels

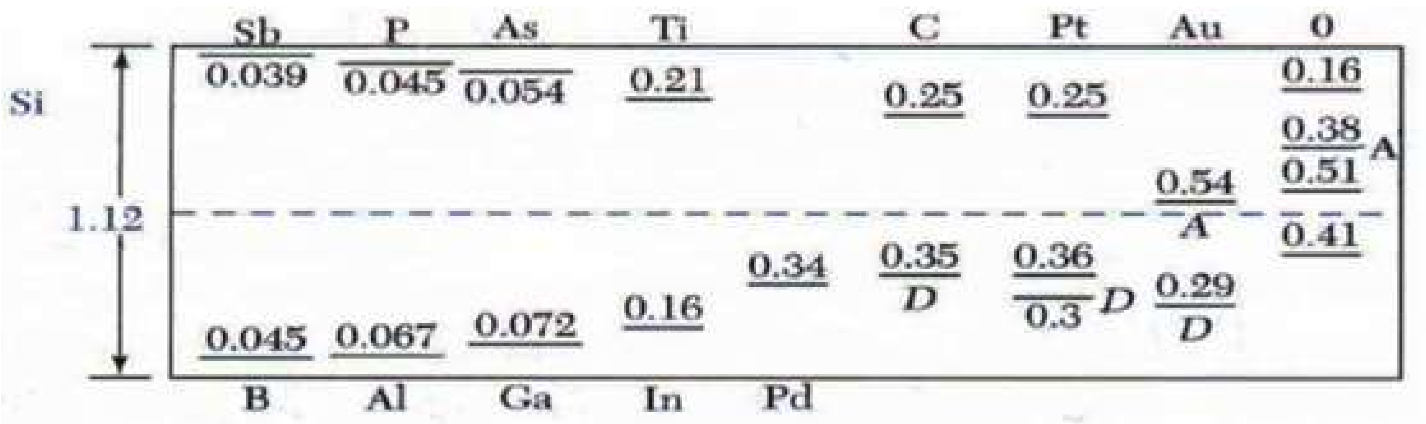


Characteristics of Deep Level Impurity

- Not easy to ionize, little effect on carrier concentration
- Multiple levels are typically generated, even producing both the donor level and the acceptor level
- It can function as a recombination center to reduce the minority carrier lifetime



Deep and shallow impurities levels



Doping in Semiconductor

If the concentration of donor impurity (e.g. phosphor) in Si is ND, the concentration of free electrons,

$$n \sim N_D$$

For Si and other semiconductors, the typical doping level are

$$N_D = 10^{15}\ \mathrm{cm}^{-3} \ldots 10^{18}\ \mathrm{cm}^{-3}$$

$$(N_D = 10^{15}\ \mathrm{cm}^{-3} \ldots 10^{18}\ \mathrm{cm}^{-3} \text{ compare to } n_i = 1.3*10^{10}\ \mathrm{cm}^{-3} \text{ in intrinsic Si})$$

$$N_D >> n_i$$

Doping provides a flexible control over semiconductor conductivity
The vast majority of microelectronic devices are based on doped semiconductors

Can the doping levels further be increased? – density of states

Doping in Semiconductor

How much would be the resistance of the (1 cm *1 cm * 1 cm) Si sample doped with donor impurities with concentration of 2*10^{16} cm^{-3}?

$$\sigma = nqu$$

$$R = \rho \frac{L}{A} = \frac{1}{\sigma} \frac{L}{A}$$

$$n = 2\times10^{16}\ \mathrm{cm}^{-3}$$

$$u_n = 1000\ \mathrm{cm}^2/(\mathrm{Vs})$$

$$q = 1.6\times10^{-19}\ \mathrm{C}$$

$$\sigma = 1.6\times10^{-19}\ \mathrm{C}\times2\times10^{16}\ \mathrm{cm}^{-3}\times1000\ \mathrm{cm}^2/(\mathrm{Vs})$$

$$= 3.2(\mathrm{Ohm}\times\mathrm{cm})^{-1}$$

$$\rho = 0.325\ \mathrm{Ohm}\times\mathrm{cm}$$

$$R = 0.325(\mathrm{Ohm}\times\mathrm{cm})\times1\ \mathrm{cm}/(1\ \mathrm{cm}\times1\ \mathrm{cm}) = 0.325\,\mathrm{Ohm}$$

Smaller than intrinsic Si in the order of 6!

The resistance of a doped Si crystal can be significantly lower
Than that of intrinsic Si

Summary

- Orbital splitting: N atoms approaching → N splitting orbitals.
- N orbital → energy gap between adjacent orbital is very small → quasi continue → energy band
- Electrons sit the orbitals with lower energy first
- Metal: energy overlaps → conduction band partially filled → conduct current under E.
- Semiconductor → small band gap → electron can jump from valence band to conduction band at RT → electron in conduction band and hole in valence band contribute to current → conductivity can be easily altered by doping
- Insulator → band gap is very large → valence band is fully filled by electrons, conduction band is empty